EXPERIMENT 6

Experiment Name: Introduction to 8255A Programmable Peripheral Interface and Experiment with Seven (7)-Segments Display

Session Objective:

- To get familiar with 8255A Interface, PIN and port configuration.
- To understand the connectivity between 8255 with I/O ports (P3 and P4), LEDs, 7-segments display and DOT Matrix units.
- To simulate an example program to display 0-9 digits in 7-segment display units.
- To simulate an example program to display characters in DOT MATRIX units.

Experiment 1: Introduction to 8255A Programmable Peripheral Interface and Experiment with Seven (7)-Segments Display and LED Connection Program

8255A is a general purpose programmable I/O device used in microprocessors. It consists of three 8-bit bidirectional I/O ports with 24 I/O pins (figure 1) which may be individually programmed in 2 groups of 12 pins and used in 3 major modes of operation.

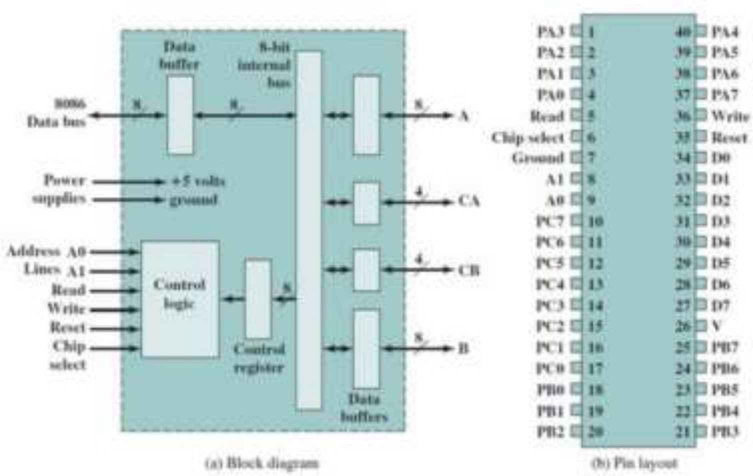


Figure 5.1: 8255 Block Diagram and PIN OUT

Carry control and status signal

A1	A0	Function
0	0	Port A
0	1	Port B
1	0	Port C
1	1	Command Register

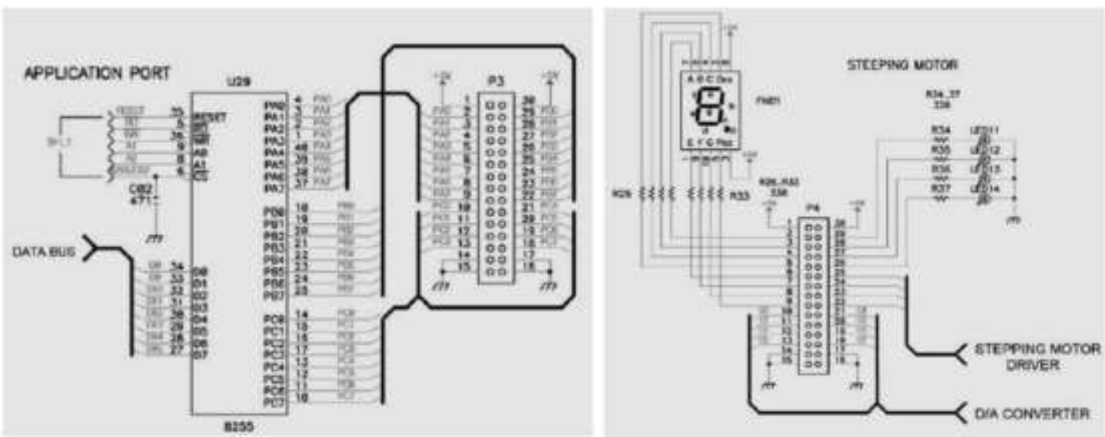


Figure 2: 82C55 interface with Microprocessor and I/O ports (P3, P4)

82C55 has three mode of operation including Mode 0, 1, 2.

Mode 0- Basic Input/Output Mode

Causes 82C55 to function either as a buffered input device the pins of Group B/Group A to be programmed as simple I/O ports.

Mode 1- Strobe Input/Output Mode

Causes operation port A and/or port B to function as latching input devices. Similar to mode 0 but data are transferred through port A/port B and handshaking (DATA READY, ACKNOWLEDGE) and interrupt request signals are provided by port C.

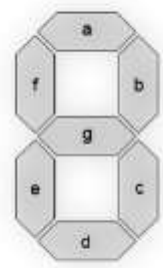
Figure 5.2: Interfacing 8255 with Seven Segment and LEDs

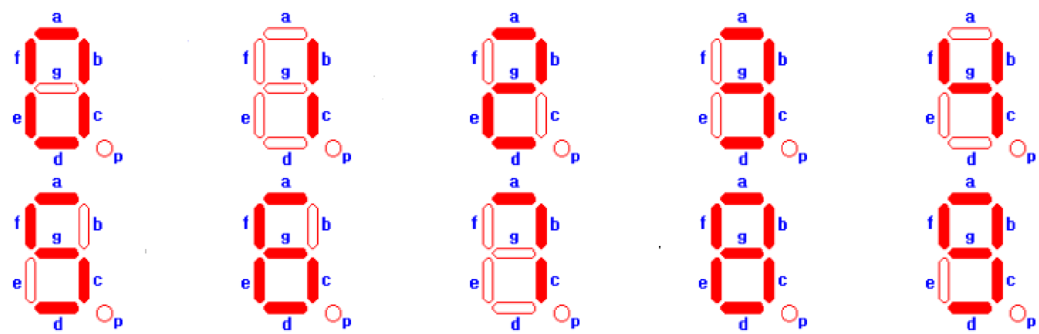
Strobe inputs signal to microprocessor retrieve data that are stored into the port registers

The address of the control register, port A, port B and port C are given below:

```
PPIC_C    EQU    1FH
PPIC      EQU    1DH
PPIB      EQU    1BH
PPIA      EQU    19H
```

Experiment 2: Write an assembly code to display 0-9 in Seven Segment Display (SSD)





- For seven segments display we use 0 for ON and 1 for OFF.
- Control register values will be the column headings of the following table:

D7	D6	D5	D4	D3	D2	D1	D0
1	0	0	0	0	0	0	0
Control Register 0- BSR mode 1- I/O mode	Mode selection for group A 00- I/O 01- Handshaking		Port A 0- Output 1- Input	Upper 4 bit of port C	Mode selection for group B 0- I/O 1- Handshaking	For port B	For lower 4 bit of port C

Assembly Code:

S SEGMENT PARA PUBLIC 'CODE'
ASSUME CS: S
ORG 1000H

START:
;control register turn on
MOV AL,80H OUT
1FH,AL

SSD:

;display 0
MOV
AL,0C0H
OUT
19H,AL
;for delay
MOV CX,0FFFFH L0:LOOP L0
;display 1
MOV
AL,0F9H
OUT
19H,AL
;for delay
MOV CX,0FFFFH L1:LOOP L1
;display 2
MOV
AL,0A4H
OUT
19H,AL
;for delay
MOV CX,0FFFFH L2:LOOP L2
;display 3
MOV
AL,0B0H
OUT
19H,AL
;for delay
MOV CX,0FFFFH L3:LOOP L3
;display 4
MOV
AL,099H
OUT
19H,AL
;for delay

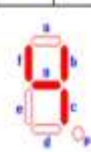
	g	f	e	d	c	b	a
1	1	0	0	0	0	0	0

	g	f	e	d	c	b	a
1	1	1	1	1	0	0	1

	g	f	e	d	c	b	a
1	0	1	0	0	1	0	0

	g	f	e	d	c	b	a
1	0	1	1	0	0	0	0

	g	f	e	d	c	b	a
1	0	0	1	1	0	0	1



	g	f	e	d	c	b	a
1	0	0	1	0	0	1	0

<u>MOV CX,0FFFFH L4:LOOP L4</u>

;display 5
MOV
AL,092H
OUT
19H,AL
;for delay

MOVCX,0FFFFH L5:LOOP L5

;display 6
MOV
AL,082H
OUT
19H,AL
;for delay

	g	f	e	d	c	b	a
1	0	0	0	0	0	1	0

MOV CX,0FFFFH L6:LOOP L6

;display 7
MOV
AL,0F8H
OUT
19H,AL
;for delay

	g	f	e	d	c	b	a
1	1	1	1	1	0	0	0

MOV CX,0FFFFH L7:LOOP L7

;display 8
MOV
AL,080H
OUT
19H,AL
;for delay

	g	f	e	d	c	b	a
1	0	0	0	0	0	0	0

MOV CX,0FFFFH L8:LOOP L8

;display 9
MOV
AL,090H
OUT
19H,AL
;for delay

	g	f	e	d	c	b	a
1	0	0	1	0	0	0	0

MOV
CX,0FFFFH
L9:LOOP L9
JMP SSD
S ENDS
END
START

Example 1: Display digits 0–9 and some characters on a 7-segment display (Kit mode).
To display the digits 0 – 9 the bit-patterns are given below.

```
P G F E D C B A
1 1 0 0  0 0 0 0 B
1 1 1 1  1 0 0 1 B
1 0 1 0  0 1 0 0 B
1 0 1 1  0 0 0 0 B
1 0 0 1  1 0 0 1 B
1 0 0 1  0 0 1 0 B
1 0 0 0  0 0 1 0 B
1 1 1 1  1 0 0 0 B
1 0 0 0  0 0 0 0 B
1 0 0 1  0 0 0 0 B
```

Task 1: Write and run the following program to display 0-9

```
B0 80      MOV AL, 10000000B
E6 1F      OUT 1F, AL           ;GOES TO CONTROL REG
B0 F0      MOV AL,11110000B
E6 1B      OUT 1B, AL           ;GOES TO PORT B
B0 00      MOV AL,00000000B
E6 1D      OUT 1D,AL            ; GOES TO PORT C

;DISPLAY STARTS HERE
LEVEL:  ;should be 100CH if the offset address of the code is 1000H
B0 C0      MOV AL, 11000000B
E6 19      OUT 19,AL            ; '0' GOES TO PORT A
B0 F9      MOV AL, 11111001B
E6 19      OUT 19,AL            ;'1'  GOES TO PORT A
B0 A4      MOV AL, 10100100B
E6 19      OUT 19,AL            ; '2' GOES TO PORT A
B0 B0      MOV AL, 10110000B
E6 19      OUT 19,AL            ; '3' GOES TO PORT A
B0 99      MOV AL, 10011001B
E6 19      OUT 19,AL            ; '4' GOES TO PORT A
B0 92      MOV AL, 10010010B
E6 19      OUT 19,AL            ; '5' GOES TO PORT A
B0 82      MOV AL, 10000010B
E6 19      OUT 19,AL            ; '6' GOES TO PORT A
```